Equation of sphere through 4 points:

$$\begin{vmatrix} x^2+y^2+z^2 & x & y & z & 1 \\ x_1^2+y_1^2+z_1^2 & x_1 & y_1 & z_1 & 1 \\ x_2^2+y_2^2+z_2^2 & x_2 & y_2 & z_2 & 1 \\ x_3^2+y_3^2+z_3^2 & x_3 & y_3 & z_3 & 1 \\ x_4^2+y_4^2+z_4^2 & x_4 & y_4 & z_4 & 1 \end{vmatrix} = 0$$

$\vec{x}_1 = \vec{a}$, $\vec{x}_2 = \vec{b}$, $\vec{x}_3 = \vec{c}$, $\vec{x}_4 = (0,0,0)$

$$\Rightarrow \begin{vmatrix} x^2+y^2+z^2 & x & y & z \\ \|\vec{a}\|^2 & a_x & a_y & a_z \\ \|\vec{b}\|^2 & b_x & b_y & b_z \\ \|\vec{c}\|^2 & c_x & c_y & c_z \end{vmatrix} = 0 \iff$$

$$\iff (x^2+y^2+z^2)\left[c_x b_y c_z + a_y b_z c_x + a_z b_x c_y - a_x b_z c_y - a_y b_x c_z - a_z b_y c_x\right] -$$

$$- x\left[\|\vec{a}\|^2(b_y c_z - b_z c_y) + \|\vec{b}\|^2(a_z c_y - a_y c_z) + \|\vec{c}\|^2(a_y b_z - a_z b_y)\right] +$$

$$+ y\left[\|\vec{a}\|^2(b_z c_x - b_x c_z) + \|\vec{b}\|^2(a_z c_x - a_x c_z) + \|\vec{c}\|^2(c_x b_z - a_z b_x)\right] -$$

$$- z\left[\|\vec{a}\|^2(b_z c_y - b_y c_x) + \|\vec{b}\|^2(a_y c_x - a_x c_y) + \|\vec{c}\|^2(a_x b_y - a_y b_x)\right] = 0$$

$$\iff (x^2+y^2+z^2)A + xB + yC + zD = 0$$

$$A = a_x(b_y c_z - b_z c_y) + a_y(b_z c_x - b_x c_z) + a_z(b_x c_y - b_y c_x)$$

$$B = \|\vec{a}\|^2(b_z c_y - b_y c_z) + \|\vec{b}\|^2(a_y c_z - a_z c_y) + \|\vec{c}\|^2(-a_y b_z + a_z b_y)$$

$$C = \|\vec{a}\|^2(b_x c_z - b_z c_x) + \|\vec{b}\|^2(a_z c_x - a_x c_z) + \|\vec{c}\|^2(a_x b_z - a_z b_x)$$

$$D = \|\vec{a}\|^2(b_y c_x - b_x c_y) + \|\vec{b}\|^2(a_x c_y - a_y c_x) + \|\vec{c}\|^2(a_y b_x - a_x b_y)$$

$$(x^2+y^2+z^2)A + Bx + Cy + Dz = 0 \iff$$

$$\iff (x^2+y^2+z^2) + \frac{Bx}{A} + \left(\frac{B}{2A}\right)^2 + \frac{Cy}{A} + \left(\frac{C}{2A}\right)^2 + \frac{Dz}{A} + \left(\frac{D}{2A}\right)^2 =$$

$$= \left(\frac{B}{2A}\right)^2 + \left(\frac{C}{2A}\right)^2 + \left(\frac{D}{2A}\right)^2 \iff$$

$$\iff \left(x + \frac{B}{2A}\right)^2 + \left(y + \frac{C}{2A}\right)^2 + \left(z + \frac{D}{2A}\right)^2 = \frac{B^2+C^2+D^2}{4A^2}$$

$$\Rightarrow \vec{n} = -\left(\frac{B}{2A}, \frac{C}{2A}, \frac{D}{2A}\right)$$

$$n = \frac{\sqrt{B^2+C^2+D^2}}{2A}$$